Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Delp, Michael Ph.D.		
Project Title:	Circulatory Remodeling with Simulated Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mdelp@fsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-214-1195
Organization Name:	Florida State University		
PI Address 1:	College of Human Sciences		
PI Address 2:	242 Sandels Building, 120 Convocation Way		
PI Web Page:			
City:	Tallahassee	**State:**	FL
Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	04/01/2001	**End Date:**	09/30/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zawieja, David (Texas A&M Health Science Center) Wilson, Emily (Texas A&M Health Science Center)		
Grant/Contract No.:	NCC 9-58-CA.002.09		
Performance Goal No.:			
Performance Goal Text:			
	The human body is exquisitely adapted for maintaining an upright posture on earth. However, when the force of gravity is removed during spaceflight, there is a cephalic fluid shift and an elimination of the head-to-foot hydrostatic pressure. This change in the fluid pressure distribution has been hypothesized to trigger adaptations within the cardiovascular system, which are subsequently rendered inappropriate upon return to the earth's gravitational environment. One of the most profound consequences of microgravity on the cardiovascular system is orthostatic intolerance. It is now becoming increasingly evident that the etiology of post-flight orthostatic intolerance is multifactorial, resulting from such factors as hypovolemia and altered regulation of the peripheral vasculature. According to Watenpaugh and Hargens (Handbook of Physiology, 1996) the primary cardiovascular adaptations that contribute to orthostatic intolerance involve the arterial, venous and lymphatic portions of the circulatory system. In order to study these phenomena on earth, the		

Task Description:	hindlimb-unloaded (HU) rat has been used to simulate the effects of microgravity. This model induces the cephalic fluid shifts, and these animals manifest many of the adaptations that are characteristic of exposure to microgravity, such as hypovolemia, a diminished capacity to elevate peripheral vascular resistance, and orthostatic hypotension. In addition, previous work with conduit and resistance arteries indicates that hindlimb unloading alters both function and structure of the arterial circulation. Therefore, using this animal model, the general aim of this proposal is to determine the effects of simulated-microgravity on 1) the molecular mechanisms mediating structural remodeling of the arterial resistance vasculature, and 2) the functional ability of the lymphatics to generate and modulate lymph flow. More specifically, we propose to: identify early regulatory events leading to hypertrophic remodeling of cerebral arteries in response to hindlimb unloading (Aim 1); characterize signaling events leading to atrophy of resistance arteries in the soleus and gastrocnemius muscle in response to hindlimb unloading (Aim 2); and evaluate the effects of hindlimb unloading on the ability of the lymphatics from different regions of the body to generate and modulate lymph flow, and thus, regulate overall body fluid homeostasis (Aim 3). These studies will provide new and important functional and mechanistic information about the etiology of microgravity-induced orthostatic intolerance. The general aim of this proposal is to determine the effects of simulated-microgravity on 1) the molecular mechanisms mediating structural remodeling of the arterial resistance vasculature, and 2) the functional ability of the lymphatics to generate and modulate lymph flow. These studies will provide new and important functional and mechanistic information about the etiology of microgravity-induced orthostatic intolerance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Orthostatic intolerance and lymphatic dysfunction are common problems among the general population. Therefore, new insights into functional vascular and lymphatic adaptation to simulated microgravity may give important insights into the mechanims that contribute to the normal and abnormal function of the arterial and lymphatic circulations. These experiments will expand our understanding of how prolonged fluid shifts, such as in a microgravity environment, affect the arterial and lymphatic circulations. The detailed study of factors associated with simulated microgravity-induced cerebral vascular smooth muscle cell hypertrophy and skeletal muscle vascular smooth muscle cell atrophy will yield essential information on the basic physiological responses of individual arterial vessels involved in blood flow and pressure regulation. In addition, the effects of cephalad fluid shifts on the ability of the lymphatics from different regions of the body to generate and modulate lymph flow, and thus, regulate overall body fluid homeostasis will be determined. This information will support the development of treatments or countermeasures to improve crew health and performance following their return to a gravitational environment.
Task Progress:	Aim 1: The initial studies were designed to determine the mRNA and protein expression of endothelial nitric oxide synthase (eNOS) in cerebral and skeletal muscle resistance arteries, which has been proposed to mediate mechanically induced vascular remodeling in other systems (Specific Aims 1 and 2). In the cerebral arteries, eNOS mRNA expression was not different between control and HU rats after 1, 14 and 28 days of hindlimb unloading. However, eNOS protein levels are significantly depressed in the middle cerebral artery (MCA) after 14 days of hindlimb unloading. The current results indicate that eNOS may be directly involved in the remodeling of the cerebral vascular hypertrophy induced by hindlimb unloading in rats. These results were presented at the Humans in Space Symposium (May 18-22, 2003), and manuscript containing these and other data relating to the remodeling of the middle cerebral artery with simulated microgravity is in press with the American Journal of Physiology: Heart and Circulatory Physiology. Aim 2: Experiments to characterize signaling events leading to atrophy of resistance arteries in the soleus and gastrocnemius muscle in response to hindlimb unloading have been completed and the results are being prepared for submission. Aim 3: Lymphatic contractile function from the cervical, thoracic duct, mesenteric and hindlimb lymphatics have been tested and characterized from control and hindlimb unloaded rats, and the results have been published or submitted for publication. The effect of hindlimb unloading was been determined to diminish the contractile function in these lymphatic vessels. More specifically, there is a 50-75% reduction in resting tone of lymphatic vessels, a 30-60% reduction in phasic contraction frequency of the lymph pump, a 60-80% reduction in the strength of phasic contractions of the lymph pump, and a significant reduction in the pressure-sensitive stimulation of the lymph pump. These results of simulated microgravity have been presented at the Bioastronautics Meeting and the Humans in Space Symposium (May 18-22, 2003), and a manuscript has been submitted and is under review at the Journal of Physiology.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Abstracts for Journals and Proceedings	Gashev AA; Zawieja DC; Delp MD "Effects of hindlimb suspension on active pumping of rat thoracic duct and mesenteric lymphatics" Experimental Biology 2003 Conference, Anaheim, CA, April 11-15, 2003. FASEB J. 2003 Apr;17:A128. , Apr-2003
Articles in Peer-reviewed Journals	Gashev AA, Davis MJ, Zawieja DC. "Inhibition of the active lymph pump by flow in rat mesenteric lymphatics and thoracic duct." J Physiol. 2002 May 1;540(Pt 3):1023-37. PMID: 11986387 , Jan-2003
Articles in Peer-reviewed Journals	Gashev AA, Davis MJ, Delp MD, Zawieja DC. "Regional variations of contractile activity in isolated rat lymphatics." Microcirculation. 2004 Sep;11(6):477-92. PMID: 15371129 , Sep-2004